

(0000CJA102120028)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Main)

AIOT

09-05-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	D	B	D	A	D	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	B	B	A	D	D	C	D	B
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	159.00	3.00	60.00	9.00	3.00	3.00	50.00	2.00	5.00	14.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	D	D	A	C	D	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	D	C	B	C	B	A	B	C
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	150.00	7.00	4.00	4.00	12.00	3.00	3.00	14.00	6.00	1.56-1.57

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	A	A	B	A	A	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	B	A	C	B	C	D	A	A
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	0.50	0.50	0.50	0.20	0.75	3.75	0.75	1.80	1.00	0.20



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. **Ans. (C)****Sol.** $\Delta V_{in} = 10\text{mV}$, $\Delta I_B = 50\mu\text{A}$, $\Delta I_C = 5\text{mA}$ $R_L = 5\text{k}\Omega$, $\Delta V_o = ?$

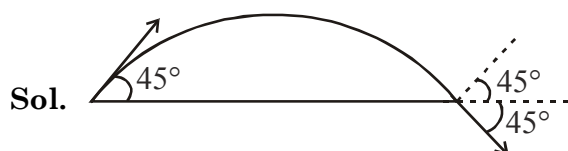
$$\frac{\Delta V_o}{\Delta V_i} = \beta \frac{R_L}{R_{in}}, \text{ where } \beta = \frac{\Delta I_C}{\Delta I_B} \text{ \& } R_{in} = \frac{\Delta V_{in}}{\Delta I_B}$$

$$\therefore \frac{\Delta V_o}{10\text{mV}} = \left(\frac{5\text{mA}}{50\mu\text{A}} \right) \left(\frac{5\text{k}\Omega \times 50\mu\text{A}}{10\text{mV}} \right)$$

$$\Rightarrow \Delta V_o = 25\text{V}$$

2. **Ans. (A)****Sol.** $H = nI = 4 \times 10^3 \text{ A/m}$

$$M = \frac{B}{\mu_0} - H = \frac{1.57}{4\pi \times 10^{-7}} - 4 \times 10^3 = 1.25 \times 10^6 \text{ A/m}$$

3. **Ans. (C)**4. **Ans. (D)****Sol.** $\Delta Q = 0$ for free expansion $\Delta W = 0$ so $\Delta U = 0$ 5. **Ans. (B)**

$$\text{Sol. } \Delta Q_{ice} (-20^\circ \rightarrow 0^\circ \text{C}) = 2 \times \frac{1}{2} \times 20 = 20 \text{ Kcal}$$

$$\Delta Q_{water} (20^\circ \rightarrow 0^\circ \text{C}) = 5 \times 1 \times 20 = 100 \text{ Kcal}$$

$\Delta Q_{water} > \Delta Q_{ice}$ so ice melts and we can see 80 Kcal of heat is remaining so 1 kg ice melts so water content 6kg.

6. **Ans. (D)****Sol.** K_0 must be COM frame.7. **Ans. (A)**

$$\text{Sol. } A = A_0 e^{-\frac{bt}{2m}}$$

$$\frac{A_1}{A_2} = e^{\frac{bT}{2m}}$$

$$\frac{bT}{2m} = \ln \left(\frac{A_1}{A_2} \right)$$

$$T = 0.479$$

$$w' = 13.11 \text{ rad/s}$$

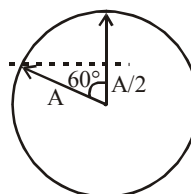
$$w^2 = w'^2 + \frac{b^2}{4m^2} = 172.88$$

$$k = mw^2 = 5 \times 10^{-3} \times 172.88 = 0.864 \text{ N/m}$$

8. **Ans. (D)****Sol.** $w = 20 \text{ rad/s}$

$$T = \frac{2\pi}{w} = \frac{\pi}{10}$$

time taken to travel 1 cm distance.



$$wt = \frac{\pi}{3}$$

$$t = \frac{\pi}{3} = \frac{\pi}{3 \times 20} = \frac{\pi}{60} = 0.052\text{s}$$

9. **Ans. (C)**

$$\text{Sol. } T^2 = \frac{4\pi^2}{GM_s} a^3$$

$$a = \frac{R_{\max} + R_{\min}}{2}$$

$$\frac{T_2^2}{T_1^2} = \left(\frac{3R}{2R} \right)^3$$

$$\frac{T_2}{T_1} = \left(\frac{3}{2} \right)^{3/2}$$

10. Ans. (A)

$$\text{Sol. } C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{2C \times 4C}{6C} = \frac{4}{3}C$$

$$Q = \frac{4C}{3} V = \frac{4 \epsilon_0 A}{3 \times 5 \times 10^{-2}} \times 100$$

$$\sigma = \frac{Q}{A} = \frac{4 \times 100 \times 100}{3 \times 5} \epsilon_0 = \frac{8000 \epsilon_0}{3}$$

$$\sigma_b = +\sigma \left(1 - \frac{1}{k_1}\right) - \sigma \left(1 - \frac{1}{k_2}\right)$$

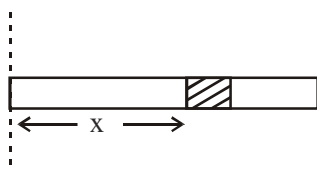
$$= -\sigma \left(\frac{1}{k_1} - \frac{1}{k_2}\right) = -\sigma \left(\frac{4-2}{4 \times 2}\right) = \frac{-2\sigma}{8} = \frac{-\sigma}{4}$$

$$= -\frac{2000 \epsilon_0}{3}$$

11. Ans. (C)

12. Ans. (B)

$$\text{Sol. } dI = [(\rho) dx] x^2$$



$$\int dI = 3 \int_0^2 x^3 dx$$

$$I = 3 \left(\frac{2^4}{4}\right) = 12 \text{ kgm}^2$$

13. Ans. (B)

$$\text{Sol. } \frac{F}{A} = Y \alpha \Delta T$$

$$= 2 \times 10^{11} \times 12.5 \times 10^{-6} \times 40$$

$$= 25 \times 40 \times 10^5 = 1 \times 10^8 \text{ N/m}^2$$

14. Ans. (B)

$$\text{Sol. } \frac{l_1}{l_2} = \frac{100}{\left(\frac{x \times 100}{100+x}\right)} = 2$$

$$100 + x = 2x$$

$$x = 100 \Omega$$

15. Ans. (A)

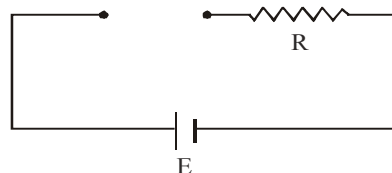
$$\text{Sol. } Q = \frac{\Delta \phi}{R} = \frac{NBA}{R}$$

$$\text{so } B = \frac{QR}{NA} = \frac{2 \times 10^{-5} \times 20}{40 \times 0.01} = 1 \times 10^{-3} \text{ T}$$

16. Ans. (D)

Sol. At $t = 0$

LR Circuit can be seen as



So

$$i = 0$$

$$P_{\text{Bat}} = 0$$

$$U_{\text{inductor}} = 0$$

$$V_L = E$$

17. Ans. (D)

$$\text{Sol. } V = V_{\text{due to } q} + V_{\text{due to induce charge at inner surface}} + V_{\text{due to change on outer surface}}$$

$$= \frac{kq}{d} + \left(-\frac{kq}{R_1}\right) + \left(\frac{kq}{R_2}\right)$$

18. Ans. (C)

$$\text{Sol. } \frac{1}{\lambda} = \frac{3R}{4} (z-1)^2$$

$$\frac{1}{\lambda} = R(z-1)^2 \left(\frac{1}{n^2} - \frac{1}{m^2}\right)$$

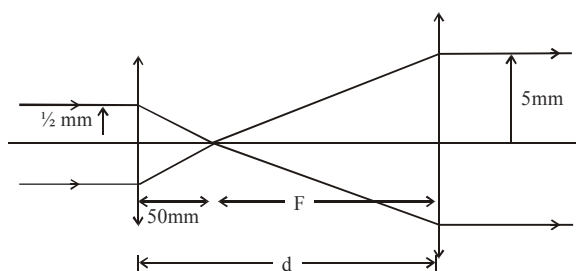
$$(z-1) = \sqrt{\frac{4}{3R\lambda}} = \sqrt{\frac{4}{3 \times 1.1 \times 10^7 \times 1.8 \times 10^{-10}}}$$

$$= 26$$

$$Z = 27$$

19. Ans. (D)

Sol.



$$\frac{1/2}{50} = \frac{5}{F}$$

$$F = 500 \text{ mm}$$

$$d = 550 \text{ mm}$$

20. Ans. (B)

Sol. Let the half life of A is t_0

$$N_A = \frac{N_{A_0}}{2^{t/t_0}} = \frac{N_{A_0}}{8}$$

$$N_B = \frac{N_{B_0}}{2^{3t_0/2}} = \frac{N_{B_0}}{64}$$

$$N_A = N_B$$

$$\frac{N_{A_0}}{8} = \frac{N_{B_0}}{64}$$

$$\frac{N_{A_0}}{N_{B_0}} = \frac{1}{8}$$

SECTION-II

1. Ans. 159.00

Sol. for 90° phase

$$X_C = R$$

$$\frac{1}{2\pi f \times 10 \times 10^{-12}} = 100 \times 10^3$$

$$f = \frac{1000}{2\pi} = 159.23 \text{ KH} \approx 159 \text{ KH}$$

2. Ans. 3.00

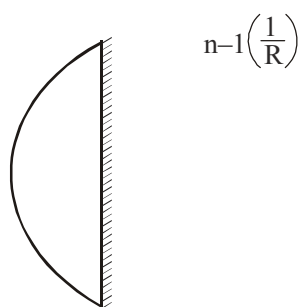
Sol. Force on dipole due to point charge is same as force on point charge due to dipole so

$$F = \frac{kP}{r^3} \times Q = \frac{kPQ}{r^3}$$

$$\text{so } N = 3$$

3. Ans. 60.00

Sol. Focal length of silvered lens.



$$\frac{1}{f_{eq}} = \frac{-2(\mu - 1)}{R}$$

$$f_{eq} = -\frac{f}{2} = -30 \text{ cm}$$

$$-\frac{1}{f_{eq}} = 2\left(\frac{n-1}{R}\right)$$

$$f_{eq} = -\frac{R}{2(n-1)}$$

and to get image of same size object must be on centre of curvature.

4. Ans. 9.00

Sol. The optical path difference developed due to insertion of two glass plates is

$$x = (\mu_2 - \mu_1)t = (1.7 - 1.4)t = 0.3t$$

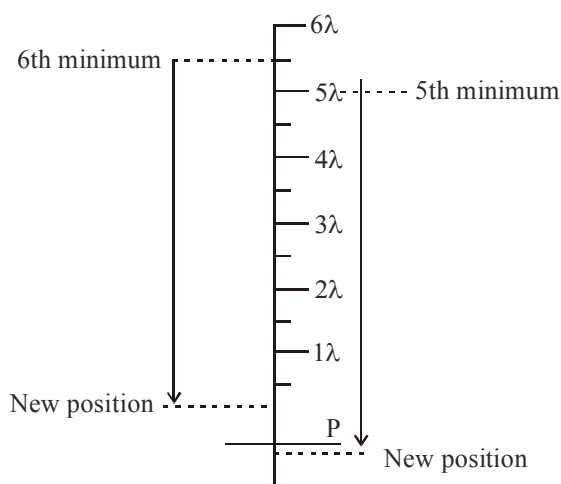
where t is the thickness of each glass plate.

$$\text{Phase difference } \delta = \frac{2\pi}{\lambda} x.$$

The intensity distribution equation is

$$I = I_0 \cos^2\left(\frac{\delta}{2}\right)$$

$$\text{At P } \frac{3}{4}I_0 = I_0 \cos^2 \frac{\delta}{2}$$



$$\text{or } \cos \frac{\delta}{2} = \pm \frac{\sqrt{3}}{2}$$

$$\text{or } \frac{\delta}{2} = n\pi \pm \frac{\pi}{6}$$

$$\text{or } \delta = 2n\pi \pm \frac{\pi}{3}$$

The corresponding path difference is

$$n\lambda \pm \frac{1}{6}\lambda.$$

After insertion of glass plates the 5th maximum goes below the point P and the 6th minimum lies above the point P. Thus due to insertion of glass plates the change of path difference is $(5 + \varepsilon)\lambda$

where $\varepsilon = 0$. Again the 6th minimum lies above P. The change in path difference is

$$\left(5 + \frac{1}{2} - \eta\right)\lambda, \eta > 0$$

$$\text{Thus, } (5 + \varepsilon)\lambda = \left(5 + \frac{1}{2} - \eta\right)\lambda$$

$$\varepsilon + \eta = \frac{1}{2} \text{ or } \varepsilon < \frac{1}{2}$$

The change of optical path must be

$$\left(5 + \frac{1}{6}\right)\lambda.$$

$$\text{Thus, } 0.3t = \left(5 + \frac{1}{6}\right)\lambda = \frac{31\lambda}{6}$$

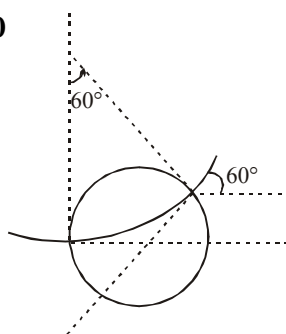
$$\text{or } t = \frac{31 \times 5400 \times 10^{-10}}{6 \times 0.3} \text{ m} = 9.3 \times 10^{-6} \text{ m}$$

5. Ans. 3.00

$$\text{Sol. } \frac{\pi}{3} = \omega t$$

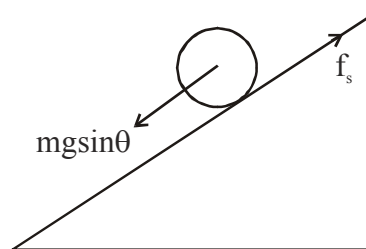
$$\frac{\pi}{3} = \omega t$$

$$t = \frac{\pi m}{3qB}$$



6. Ans. 3.00

$$\text{Sol. } mg \sin \theta - f_s = ma$$



$$f_s R = \frac{mR^2}{2} \alpha$$

$$a = R\alpha$$

$$a = \frac{2g \sin \theta}{3}$$

$$= \frac{2 \times 10 \times \frac{1}{2}}{3} = \frac{10}{3} \text{ m/s}^2$$

7. Ans. 50.00

Sol. For minimum strain on eye final image must be at infinite.

$$v_f \rightarrow \infty \text{ so } u_e = f_e = 5 \text{ cm}$$

$$\text{so } v_o \text{ (image distance of objective)} = 5 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \text{ for objective}$$

$$\frac{1}{+5} - \frac{1}{u_o} = \frac{1}{1}$$

$$\frac{1}{u_o} = \frac{1}{5} - 1 = -\frac{4}{5}$$

$$u_o = -\frac{5}{4}$$

$$u_o = -\frac{50}{40} \text{ cm}$$

$$n = 50 \text{ cm}$$

8. Ans. 2.00

$$\text{Sol. } a = \frac{\rho \frac{V}{2} g + 2\rho \frac{V}{2} g - \rho v g}{\rho V} = \frac{g}{2}$$

$$K = 2$$

9. Ans. 5.00

Sol. $T = 0.5 \text{ N}$

$$\mu = \frac{m}{l} = 5 \times 10^{-3} \text{ kg/m}$$

$$v = \sqrt{\frac{T}{\mu}} = 10 \text{ m/s}$$

$$\lambda = \frac{v}{f} = 0.1 \text{ m} = 10 \text{ cm}$$

distance between two successive nodes

$$\frac{\lambda}{2} = 5 \text{ cm}$$

10. Ans. 14.00

$$\text{Sol. } V_{PQ} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80}{0.06/4.8}} = 80 \text{ m/s}$$

$$V_{QR} = \sqrt{\frac{80}{0.2/2.56}} = 32 \text{ m/s}$$

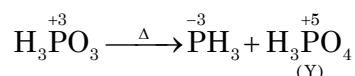
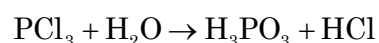
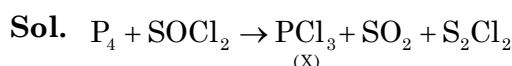
$$t = \frac{4.8}{80} + \frac{2.56}{32} = 0.14 \text{ s}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)

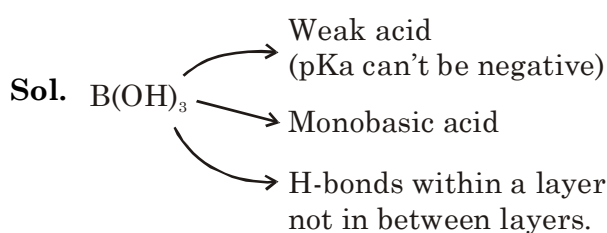


2. Ans.(A)

3. Ans.(D)

Sol. Informational

4. Ans.(D)



5. Ans.(D)

Sol. 2° amines form N-nitrosoamine upon treatment with $NaNO_2$ & HCl .

6. Ans.(A)

Sol. $Z = \frac{PV}{nRT}$

$$V = \frac{0.5 \times 0.0821 \times 273}{100}$$

$$V = 0.112 \text{ litres}$$

$$\left(P + \frac{a}{(V)^2}\right)(V - b) = RT$$

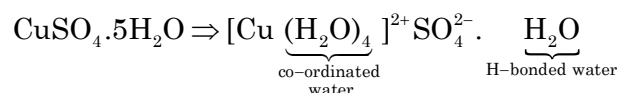
$$a = \left(1 - \frac{PV}{RT}\right)VRT$$

$$a = (1 - 0.5) \times 0.112 \times 0.0821 \times 273$$

$$a = 1.254 \text{ atm L}^2 \text{ mol}^{-2}$$

7. Ans.(C)

Sol. $BaCl_2 \cdot 2H_2O$: interstitial water
 $[Cr(H_2O)_6]Cl_3$: co-ordinated water



8. Ans.(D)

9. Ans.(A)

Sol. $\Delta T_f = i \times k_f \times m$

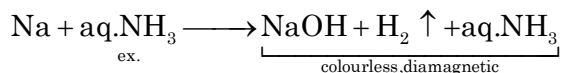
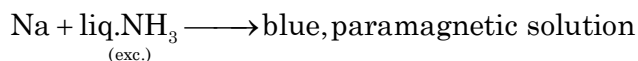
$$2 = i \times 1.72 \times \frac{\frac{20}{50}}{1000}$$

$$2 = i \times 1.72 \times \frac{20}{172} \times \frac{1000}{50}$$

$$i = \frac{1}{2}$$

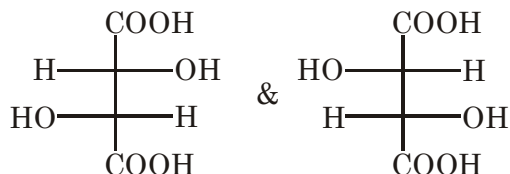
10. Ans.(C)

Sol.



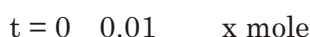
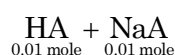
11. Ans.(B)

Sol. Optically active tartaric acids are-



12. Ans.(A)

Sol. In 0.1 L buffer solution



$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$5.5 = 5 + \log \frac{0.01 + x}{0.01 - x}$$

$$0.5 = \frac{\log 0.01 + x}{0.01 - x}$$

$$3.16 = \frac{0.01 + x}{0.01 - x}$$

$$x = \frac{2.16}{4.16} \times 10^{-2}$$

$$\frac{w}{40} = \frac{2.16}{4.16} \times 10^{-2}$$

$$W = 2.08 \times 10^{-1} \text{ g}$$

13. Ans.(D)

Sol. Ph : C_6H_5^- : Phenyl



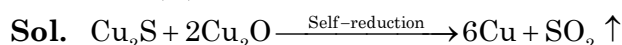
ox. state of central atom = +2

14. Ans.(C)

Sol. Given reaction in Kolbe's electrolysis

15. Ans.(B)

16. Ans.(C)



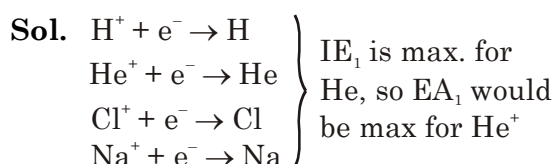
17. Ans.(B)

Sol. Given reaction is an example of intramolecular Cannizzaro reaction.

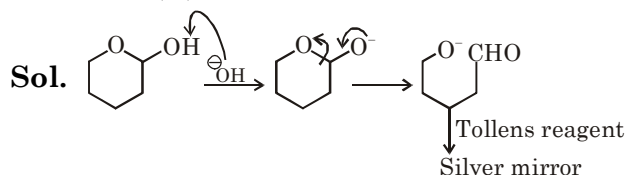
18. Ans.(A)

Sol. Fact based

19. Ans.(B)

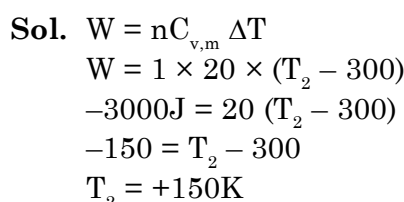


20. Ans.(C)

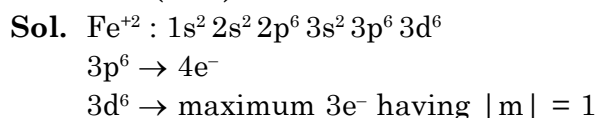


SECTION-II

1. Ans.(150.00)

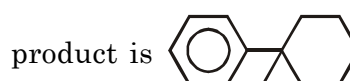


2. Ans.(7.00)

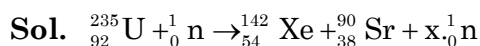


3. Ans.(4.00)

Sol. $x = 1, y = 3$



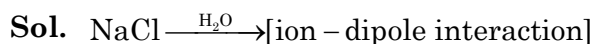
4. Ans.(4.00)



conserve mass no. $\rightarrow 235 + 1 = 142 + 90 + x(1)$

$$x = 4$$

5. Ans.(12.00)



$$\text{interaction energy} \propto \frac{1}{d^2}$$

$$\left[\text{interaction force} \propto \frac{1}{d^3} \right]$$

6. Ans.(3.00)

Sol. Amino acid Number of chiral carbons

Glycine 0

Alanine 1

Arginine 1

lysine 1

Total 3

7. Ans.(3.00)

Sol. $t_{1/2} \propto \frac{1}{(P)^{n-1}}$ where P $\rightarrow$ partial pressure

$$\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{P_2}{P_1} \right)^{n-1}$$

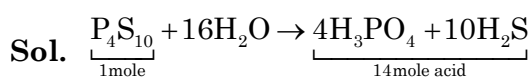
$$\frac{235}{940} = \left(\frac{250}{500} \right)^{n-1}$$

$$\frac{1}{4} = \left(\frac{1}{2} \right)^{n-1}$$

$$\left(\frac{1}{2} \right)^2 = \left(\frac{1}{2} \right)^{n-1}$$

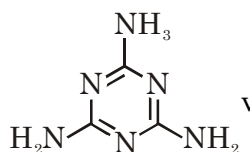
$$n = 3$$

8. Ans.(14.00)



9. Ans.(6.00)

Sol. Structure of melamine is



10. Ans.(1.56 - 1.57)

Sol. $E = 1.67 - \frac{0.059}{4} \log Q$

$$Q = \frac{[\text{Fe}^{2+}]^2}{(P_{\text{O}_2}(\text{g}))[\text{H}^+(\text{g})]^4}$$

$$Q = \frac{[10^{-3}]^2}{(0.1)(10^{-3})^4} = (10)^7$$

$$E = 1.67 - \frac{0.059}{4} \log(10)^7$$

$$E = 1.56675 \text{ V}$$

SECTION-I

1. Ans. (B)

Sol. $p \rightarrow (q \rightarrow p) \equiv \sim p \vee (q \rightarrow p) = \sim p \vee (\sim q \vee p)$

Since $p \vee \sim p$ is always true

$$= \sim p \vee p \vee q = p \rightarrow (p \vee q)$$

2. Ans. (D)

Sol. The given lines are

$$x - 1 = \frac{y + 3}{-\lambda} = \frac{z - 1}{\lambda} = s \quad (1)$$

$$\text{and } 2x = y - 1 = \frac{z - 2}{-1} = t \quad (2)$$

The lines are coplanar,

$$\text{if } \begin{vmatrix} 0 - (-1) & -1 - 3 & -2 - (-1) \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix} = 0$$

Operate $C_2 \rightarrow C_2 + C_3$

$$\begin{vmatrix} 1 & -5 & -1 \\ 1 & 0 & \lambda \\ \frac{1}{2} & 0 & -1 \end{vmatrix} = 0 \Rightarrow 5 \left(-1 - \frac{\lambda}{2} \right) = 0 \Rightarrow \lambda = -2$$

3. Ans. (A)

Sol. $(i - j) \cdot [(j - k) \times (k - i)]$
 $= (i - j) \cdot (j \times k - j \times i - k \times k + k \times i)$
 $= (i - j) \cdot (i + k - 0 + j)$
 $= i \cdot i + i \cdot k + i \cdot j - j \cdot i - j \cdot k - j \cdot j$
 $= i^2 + 0 + 0 - 0 - 0 - j^2 = i^2 - j^2 = 1 - 1$
 $= 0$

4. Ans. (A)

Sol. Let $(x_1, y_1) \equiv (at_1^2, 2at_1)$

and $(x_2, y_2) \equiv (at_2^2, 2at_2)$.

Then, $(x_3, y_3) = [at_1t_2, a(t_1 + t_2)]$

$$\therefore x_1x_2 = at_1^2 \cdot at_2^2 = (at_1t_2)^2 = x_3^2$$

$$\text{and } y_3 = a(t_1 + t_2) = \frac{1}{2}(y_1 + y_2)$$

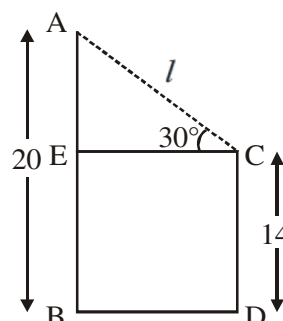
$\Rightarrow x_1, x_3, x_2$ are in G.P. and y_1, y_2, y_3 are in A.P.

5. Ans. (A)

Sol. Let $AB = 20$ and $CD = 14$ be the two poles.

$AC = l$ is the length of the wire.

CE , the horizontal through C , $\angle ACE = 30^\circ$



$$\frac{AE}{l} = \sin 30^\circ = \frac{1}{2}$$

$$\therefore l = 2AE = 12m$$

6. Ans. (B)

Sol. We have, $\sum_{r=0}^{10} {}^{20}C_r = {}^{20}C_0 + {}^{20}C_1 + \dots + {}^{20}C_{10}$

$$\text{But } {}^{20}C_0 + {}^{20}C_1 + \dots + {}^{20}C_{20} = 2^{20}$$

$$\therefore \sum_{r=0}^{10} {}^{20}C_r = ({}^{20}C_0 + {}^{20}C_1 + \dots + {}^{20}C_{20}) - ({}^{20}C_{11} + {}^{20}C_{12} + \dots + {}^{20}C_{20})$$

$$= 2^{20} + {}^{20}C_{10} - ({}^{20}C_{10} + {}^{20}C_9 + \dots + {}^{20}C_0)$$

$$\Rightarrow 2({}^{20}C_0 + {}^{20}C_1 + \dots + {}^{20}C_{10}) = 2^{20} + {}^{20}C_{10}$$

$$\therefore {}^{20}C_0 + {}^{20}C_1 + \dots + {}^{20}C_{10} = 2^{19} + \frac{1}{2} {}^{20}C_{10}$$

7. Ans. (A)

Sol. We have,

$$f'(x) = e^{x(1-x)} + x e^{x(1-x)} \cdot (1-2x)$$

$$= e^{x(1-x)} [1 + x(1-2x)]$$

$$= e^{x(1-x)} \cdot (-2x^2 + x + 1)$$

Now, the sign-scheme for $-2x^2 + x + 1$ is

$$\begin{array}{ccccccc} & & + & & + & & \\ & & | & & | & & \\ -ve & -1/2 & & +ve & 1 & & -ve \end{array}$$

$\therefore f'(x) \geq 0$ if $x \in \left[-\frac{1}{2}, 1\right]$ $e^{x(1-x)}$ is always positive

So, $f(x)$ is increasing on $\left[-\frac{1}{2}, 1\right]$

8. **Ans. (A)**

Sol. We have,

$$(xy^2 + 2x^2y^3) dx + (x^2y - x^3y^2) dy = 0$$

$$\Rightarrow xy(y dx + x dy) + x^2y^2(2y dx - x dy) = 0$$

$$\Rightarrow \frac{d(xy)}{x^2y^2} + \left(\frac{2}{x} dx - \frac{1}{y} dy \right) = 0$$

(Dividing by x^3y^3)

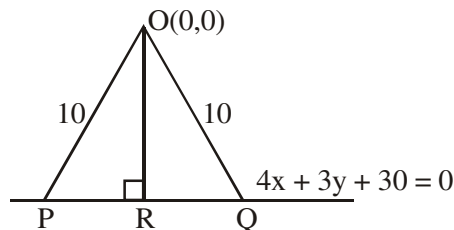
On integrating,

$$\text{we get } -\frac{1}{xy} + 2\log|x| - \log|y| = c$$

9. **Ans. (A)**

Sol. Let $OR \perp PQ$

$$\text{Then, } OR = \frac{|4(0) + 3(0) + 30|}{\sqrt{16+9}} = \frac{30}{5} = 6.$$



$$\therefore PR = \sqrt{OP^2 - OR^2} = \sqrt{100 - 36}$$

$$\text{and } PQ = 2PR = 16$$

$\therefore$ Area of

$$\Delta OPQ = \frac{1}{2} \times PQ \times OR = \frac{1}{2} \times 16 \times 6 = 48$$

10. **Ans. (C)**

Sol. $f(0) = a, f(1) = a + b + c, f\left(\frac{1}{2}\right) = a + \frac{b}{2} + \frac{c}{4}$

$$\therefore I = \int_0^1 f(x) dx = \int_0^1 (a + bx + cx^2) dx$$

$$= \left(ax + \frac{bx^2}{2} + \frac{cx^3}{3} \right)_0^1 = \frac{1}{6} (6a + 3b + 2c)$$

$$= \frac{1}{6} \left[f(0) + 4f\left(\frac{1}{2}\right) + f(1) \right]$$

11. **Ans. (B)**

Sol. Given that

$$x^{2021} + \frac{1}{x^{2021}} + \tan y + \cot y = 4$$

$$\Rightarrow x = 1 \text{ and } y = \frac{\pi}{4} \Rightarrow \tan y = 1$$

$$\Rightarrow \frac{2 \tan \frac{y}{2}}{1 - \tan^2 \frac{y}{2}} = 1 \Rightarrow \tan^2 \frac{y}{2} + 2 \tan \frac{y}{2} - 1 = 0$$

12. **Ans. (D)**

Sol. Given statement $r = \sim p \leftrightarrow q$

Statement-1 : $r_1 = (p \wedge \sim q) \vee (\sim p \wedge q)$

Statement-2 : $r_2 = \sim(p \leftrightarrow q) = (p \wedge q) \vee (\sim p \wedge \sim q)$

From the truth table of r, r_1 and $r_2, r_1 = r_2$.
Hence Statement-1 is true and Statement-2 is false

13. **Ans. (B)**

Sol. We have $\sqrt{x} + \sqrt{y} = 4$ (1)

$$\Rightarrow \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0, \text{ or } \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$$

Since the tangent is equally inclined to the axes,

$$\frac{dy}{dx} = \tan 45^\circ \text{ or } \tan 135^\circ \text{ i.e., } 1 \text{ or } -1.$$

$$\text{Thus } -\frac{\sqrt{y}}{\sqrt{x}} = \pm 1. \text{ This gives } y = x$$

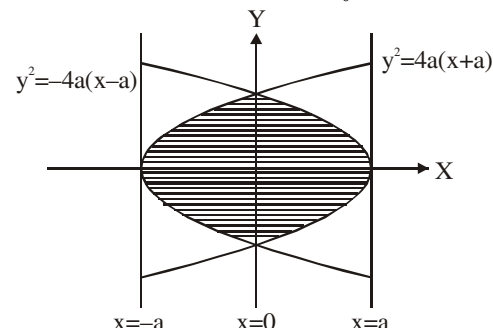
$$\text{From (1), } \sqrt{x} + \sqrt{x} = 4 \Rightarrow x = 4.$$

$$\text{Also, } y = x = 4.$$

The point is (4, 4)

14. **Ans. (A)**

Sol. The required area = $4 \int_0^a \sqrt{4a(a-x)} dx$



$$= 4.2\sqrt{a} \cdot \left[\frac{-2(a-x)^{3/2}}{3} \right]_0^a = \frac{16}{3} a^2$$

The correct option is (A)

15. Ans. (C)

Sol. We have

$$\frac{dy}{dx} = y \tan x - y^2 \sec x$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \tan x = -\sec x$$

$$\text{Putting } \frac{1}{y} = V \Rightarrow \frac{-1}{y^2} \frac{dy}{dx} = \frac{dV}{dx},$$

we obtain $\frac{dV}{dx} + \tan x \cdot V = \sec x$, which is linear

$$\text{I.F.} = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

Hence, the solution is

$$V \sec x = \int \sec^2 x dx + c \text{ or } \frac{1}{y} \sec x = \tan x + c$$

$$\text{or } \sec x = y(c + \tan x)$$

16. Ans. (B)

$$\text{Sol. } \int_0^3 [\sqrt{x}] dx = \int_0^1 [\sqrt{x}] dx + \int_1^3 [\sqrt{x}] dx$$

$$\left[\because [\sqrt{x}] = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 3 \end{cases} \right]$$

$$= \int_0^1 0 dx + \int_1^3 1 dx = x \Big|_1^3 = 2$$

17. Ans. (C)

$$\text{Sol. } A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^8 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

$$\text{Similarly, } A^{64} = \begin{bmatrix} 1 & 32 \\ 0 & 1 \end{bmatrix}$$

18. Ans. (D)

Sol. It is in A.P. for which $a = 20$, $d = 2 \frac{2}{3} = \frac{8}{3}$

$$\text{Now, } S_n > 1568$$

$$\Rightarrow \frac{n}{2} \left[40 + (n-1) \frac{8}{3} \right] > 1568$$

$$\Rightarrow \frac{n}{2} \times \frac{112 + 8n}{3} > 1568$$

$$\Rightarrow n^2 + 14n > \frac{6}{8} \times 1568 = 1176$$

$$\Rightarrow n^2 + 14n - 1176 > 0,$$

$$\text{or } (n + 42)(n - 28) > 0$$

As n is positive, $n - 28 > 0$ i.e., $n > 28$

$\therefore$ Minimum value of $n = 29$.

19. Ans. (A)

Sol. For a non-trivial solution, we must have,

$$\begin{vmatrix} 1 & 4a & a \\ 1 & 3b & b \\ 1 & 2c & c \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 4a & a \\ 0 & 3b-4a & b-a \\ 0 & 2c-4a & c-a \end{vmatrix}$$

$$(\text{Applying } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1)$$

$$\Rightarrow (3b - 4a)(c - a) - (2c - 4a)(b - a) = 0$$

$$\Rightarrow bc + ab - 2ac = 0 \Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

20. Ans. (A)

Sol. Required probability

$$= \frac{{}^7C_3}{{}^8C_4} = \frac{\frac{7 \times 6 \times 5}{1 \times 2 \times 3}}{\frac{8 \times 7 \times 6 \times 5}{1 \times 2 \times 3 \times 4}} = \frac{4}{8} = \frac{1}{2}$$

SECTION-II

1. Ans. 0.50

$$\text{Sol. } \int_0^1 (1+x)^{10} dx = \int_0^1 (C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots + C_{10} x^{10}) dx$$

$$\Rightarrow \frac{C_0}{1} + \frac{C_1}{2} + \dots + \frac{C_{10}}{11} = \left(\frac{(1+x)^{11}}{11} \right)_0^1 = \frac{2^{11} - 1}{11}$$

2. Ans. 0.50

Sol. Since $z = 2$

$$\Rightarrow \frac{x-1}{2} = \frac{y-2}{3} = \frac{-1}{4} \Rightarrow x = \frac{1}{2}, y = \frac{5}{4}$$

Since it satisfy $\frac{x^2}{a^2} - \frac{y^2}{1} = 1$,

$$\therefore \frac{1}{4a^2} - \frac{25}{16} = 1 \Rightarrow \frac{1}{4a^2} = \frac{41}{16} \Rightarrow a^2 = \frac{4}{41}$$

3. Ans. 0.50

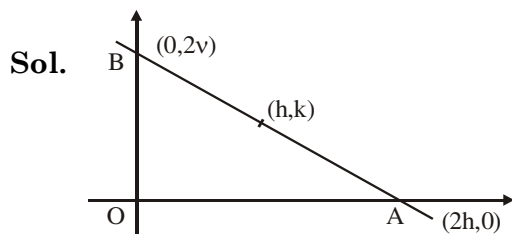
Sol. $f(x) = \frac{\alpha_x}{x+1}$

$$f(f(x)) = x$$

$$\alpha^2 = \alpha x + x + 1 \Rightarrow \alpha = -1$$

$$\frac{\alpha^2}{2} = \frac{1}{2}$$

4. Ans. 0.20

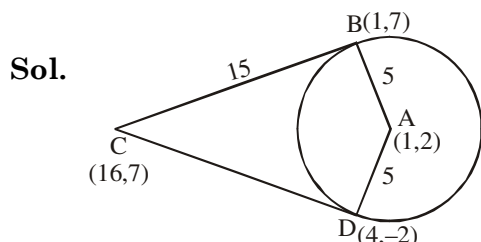


$$AB = 10$$

$$4h^2 + 4k^2 = 100$$

$$x^2 + y^2 = 25$$

5. Ans. 0.75



$$\text{Area of } ABCD = 2\Delta ABC = 75$$

$$\frac{\Delta}{100} = \frac{3}{4}$$

6. Ans. 3.75

Sol. $\frac{x_1 + x_2 + \dots + x_{10}}{10} = 4.5 \Rightarrow x_1 + x_2 + \dots + x_{10} = 45$

Also,

$$\frac{x_{11} + x_{12} + \dots + x_{40}}{30} = 3.5 \Rightarrow x_{11} + x_{12} + \dots + x_{40} = 105$$

$$\therefore x_1 + x_2 + \dots + x_{40} = 150$$

$$\therefore \frac{x_1 + x_2 + \dots + x_{40}}{40} = \frac{150}{40} = \frac{15}{4}$$

7. Ans. 0.75

Sol. $s_n = cn^2$

$$t_n = s_n - s_{n-1} = c(n^2 - (n-1)^2) = c[2n - 1]$$

$$\text{Required sum} = c^2 \sum (2n-1)^2$$

$$= c^2 \sum (4n^2 - 4n + 1) = n \left(\frac{4n^2 - 1}{3} \right) c^2$$

$$p = 4, s = 3$$

8. Ans. 1.80

Sol. $n = 6^4$

$$\frac{n}{6!} = \frac{6^4}{6!} = 1.80$$

9. Ans. 1.00

Sol. $x_n = e^{\frac{1}{2n}\pi}$

$$x_1 x_2 x_3 \dots = e^{i\left(\frac{\pi}{2} + \frac{\pi}{2^2} + \dots\right)} = e^{i\left(\frac{\pi/2}{1-1/2}\right)} = e^{i\pi} = -1$$

10. Ans. 0.20

Sol. $f(2) < 0 \Rightarrow 4 - 2(k+1) + k^2 + k - 8 < 0$

$$k^2 - k - 6 < 0 \Rightarrow k \in (-2, 3)$$